Journal of Food Composition and Analysis 147 (2025) 108076

Contents lists available at ScienceDirect

Journal of Food Composition and Analysis

journal homepage: www.elsevier.com/locate/jfca

Aroma characterization and reconstitution of strong-aroma baijiu: Insights into sensory profiles and key odor-active compounds from Chinese core production areas

Yue MA [a,b,c,d,1], Ran Xiangli [a,b,1], Dong Zhao [c,d,*], Jia Zheng [c,d], Jian SU [c,d], Yan XU [a,b,**]

[a] Laboratory of Brewing Microbiology and Applied Enzymology, School of Biotechnology and Key Laboratory of Industrial Biotechnology of Ministry of Education, Jiangnan University, Wuxi, Jiangsu, PR China
[b] China Key Laboratory of Microbiomics and Eco-Brewing Technology for Light Industry, Wuxi, Jiangsu, PR China
[c] Wuliangye Yibin Co., Ltd, Yibin, Sichuan 644000, PR China
[d] Sichuan Key Laboratory of Solid-State Fermentation Resources Utilization, Ltd, Yibin, Sichuan 644000, PR China

ARTICLE INFO

Keywords:
Chinese liquor
Sensory evaluation
Odorants
Flavor
Aroma recombination

ABSTRACT

Strong-aroma baijiu, a globally significant distilled spirit, is renowned for its complex aroma profile. This study employed sensory analysis and advanced analytical techniques to characterize the aroma profiles of ten commercial baijiu samples from different regions and alcohol levels. A total of 97 odor-active compounds were identified, with 26 key compounds, including nonanal, 2-ethyl-6-methylpyrazine, hexan-1-ol, hexanoic acid, octanoic acid, ethyl pentanoate, playing crucial roles in aroma differentiation. Partial least square-discriminant analysis (PLS-DA) effectively classified samples into sensory groups, and aroma reconstitution experiments confirmed the contribution of 27 compounds to the overall aroma profile. These findings provide valuable insights into the chemical and sensory foundations of strong-aroma baijiu, offering practical guidance for quality control and product development in the baijiu industry.

1. Introduction

Baijiu is one of the world's most consumed alcoholic beverages and currently leads the global spirits market with an annual revenue of over 161 billion U.S. dollars, according to the data recorded in 2023 (Ou, 2025a). Strong-aroma baijiu (or nongxiang baijiu in Chinese) is one of the most popular flavor types of baijiu and has held the first position in the market for almost 30 years, and in 2023, it still accounted for about 50 percent of the market (Ou, 2025b). This type of baijiu is characterized by its complex fermentation process, which involves the use of a unique mixture of grains, including sorghum, wheat, and rice, and fermentation in mud pits (Fang & Xu, 2023). The mixture of grains will be fermented for several months and then will be mixed with rice husk and distilled under solid-state conditions. The fresh distilled baijiu is then stored and aged in a pottery for several months to years, and all these processes contribute to a rich and diverse sensory profile of a strong-aroma baijiu (Jin et al., 2017; Wang et al., 2022).

Consumer preference for a product is influenced by its intense and lasting aroma (Paluchová et al., 2017). The most typical aroma characteristics for strong-aroma baijiu given by industrial experts is the "cellar" (or "Jiao" in Chinese) attribute (Qiao et al., 2023; Wang et al., 2022), which should positively correlate with "pineapple", "honey" and "alcohol" odors (Xiangli et al., 2024). In addition, lexicons, such as "floral", "fruity", "grain", "acidic", "grass" and "toasted" have been generated by other researchers to describe strong-aroma baijiu (Chen et al., 2024; Hong et al., 2023). Research on aroma compounds in strong-aroma baijiu has been conducted for 20 years (Fan & Qian, 2005, 2006), and by 2022, 861 trace volatile components have been identified

* Corresponding author at: Wuliangye Yibin Co., Ltd, Yibin, Sichuan 644000, PR China.
** Corresponding author at: Laboratory of Brewing Microbiology and Applied Enzymology, School of Biotechnology and Key Laboratory of Industrial Biotechnology of Ministry of Education, Jiangnan University, Wuxi, Jiangsu, PR China.
E-mail addresses: yuema@jiangnan.edu.cn (Y. MA), 6220210030@stu.jiangnan.edu.cn (R. Xiangli), 33321523@qq.com (D. Zhao), zhengwanqi86@163.com (J. Zheng), 232666600@qq.com (J. SU), yxu@jiangnan.edu.cn (Y. XU).
[1] These authors contributed equally to this work and should be considered co-first authors

https://doi.org/10.1016/j.jfca.2025.108076
Received 14 April 2025; Received in revised form 22 July 2025; Accepted 22 July 2025
Available online 23 July 2025
0889-1575/© 2025 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

in strong-aroma baijiu (Wang et al., 2022). Ethyl hexanoate was identified as the most typical and important aroma compound (Fan & Qian, 2006; He et al., 2021; Zhao & Zheng, 2019; Zhao et al., 2018), and it has been used as an indicator compound to determine whether a baijiu is a strong-aroma baijiu according to the Chinese national standard (GB/T 15109–2021). In the last decade, a number of studies have used or partially used the Sensomics approach to analyze the key aroma compounds in different brands of strong-aroma baijiu (Chen et al., 2024; Granvogl & Schieberle, 2022; Hong et al., 2023; Zhao et al., 2018). Their studies proved the contribution of ethyl hexanoate, ethyl butanoate, ethyl acetate, ethyl octanoate, ethyl pentanoate, ethyl 3-phenylpropanoate, hexanoic acid, and butanoic acid to the overall aroma profile of strong-aroma baijiu (Chen et al., 2024; Hong et al., 2023; Zhao et al., 2018). Even though the key aroma compounds in strong-aroma baijiu have been clarified and most of them were the same, previous studies showed that slight changes in the concentration of some compounds can significantly affect the aroma perception of an alcoholic beverage (Ma et al., 2023; Ma et al., 2025). Based on that, it is not surprising that strong-aroma baijiu from different brands have significant differences in their overall aroma perception, since samples from different brands have significant differences in the concentration of these key aroma compounds (He et al., 2020; Hong et al., 2021).

Like most distilled spirits, baijiu needs to be blended with a variety of raw or base baijiu to form a product that meets the quality needs of the brand before it becomes an official commodity, and one of the major criteria for judging the quality of a product is the aroma characteristics of the baijiu, which are normally evaluated by national judges. Therefore, it is crucial for technicians in the baijiu industry to understand how to combine raw baijiu with different aroma characteristics into commercial baijiu with stable aroma quality, and this part of the blending knowledge is also a core and bottleneck technology. For a raw strong-aroma baijiu, its aroma style is normally divided by production regions into Sichuan region style and Jianghuai region style, and strong-aroma baijiu from Sichuan region style can be further divided by raw materials for brewing into five-grains style, represented by Wuliangye, and single-grain style, represented by Luzhou Laojiao (Fang & Xu, 2023). There have been some studies analyzing the differences in the content of some key aroma compounds in raw and commercial strong-aroma baijiu from different regions (He et al., 2020; Hong et al., 2021), however, the contribution of these compounds to the differentiation of the aroma profile of the samples has never been verified by aroma reconstitution experiments. In addition, for researchers or technicians involved in the aroma blending process, it should be more instructive to use aroma profile differences, as opposed to regional differences, as a criterion for screening and comparing differences in target compounds, but this part of the study is currently little covered.

To address these gaps, this study represents a systematic analysis of the aroma characteristics of representative strong-aroma baijiu samples from various brands and alcohol levels, employing a combination of advanced analytical techniques and sensory evaluations. By integrating gas chromatography–olfactometry (GC–O), comprehensive two-dimensional gas chromatography coupled with time-of-flight mass spectrometry (GC×GC–TOFMS), and aroma reconstitution experiments, the research aims to provide new insights into the chemical basis and sensory differentiation of strong-aroma baijiu. Furthermore, this study is the first to systematically compare the aroma profiles of both high- and low-alcohol samples, highlighting the significance of the relative proportions of compounds in aroma perception.

2. Material and methods

2.1. Baijiu samples

Ten commercial baijiu samples of different alcohol content were provided by five different baijiu factories located in the different strong-aroma baijiu core production regions. These samples were selected by experts from the China Alcoholic Drinks Association to ensure that they were typical of the strong-aroma styles of baijiu from these different core production regions. All samples were stored thermostatically in the dark. The information about these samples is given in Table 1.

2.2. Chemicals

Absolute ethanol (≥99.8 %, HPLC grade), dichloromethane (≥99.8 %, HPLC grade), methanol (≥99.9 %, HPLC grade), ethyl acetate (≥99.8 %, HPLC grade), and acetone (≥99.8 %, HPLC grade) were purchased from Sigma Aldrich Trading Co., Ltd. (Shanghai, China). Analytical-grade anhydrous sodium sulfate (Na_2SO_4) and sodium chloride (NaCl) were purchased from China National Pharmaceutical Group Corp. (Shanghai, China). All chemical standards involved for quantification (see Supplementary Table 1) were chromatographic grade with purity≥ 97 % and were obtained from Sigma Aldrich Trading Co., Ltd. (Shanghai, China). All chemical standards involved for aroma reconstitution (see Table 3) were food grade with purity≥ 95 % and were obtained from Sigma Aldrich Trading Co., Ltd. (Shanghai, China). Ultrapure water was obtained from a Milli-Q purification system (Millipore, Bedford, MA, USA).

2.3. Aroma extraction by liquid–liquid extraction

Liquid–liquid extraction, which has been successfully used in the aroma extraction of baijiu samples, was used for five representative strong aroma baijiu samples to screen potential odor-active compounds by gas chromatography-olfactometry (GC-O) through human “sniffing” detection. For each baijiu sample, 50 mL of sample was diluted to 10 % alcohol by volume (ABV) with saturated sodium chloride solution, and the sample was placed in a separatory funnel. The diluted baijiu sample was then extracted three times with 50 mL of redistilled dichloromethane. The three extracts were collected and overnight dried with anhydrous Na_2SO_4 and concentrated to a final volume of 500 μL under a gentle stream of nitrogen and then stored at −20°C until analysis. All extractions were carried out in triplicate.

2.4. Gas chromatography–olfactometry (GC–O) and gas chromatography mass spectrometric (GC–MS) analysis

GC-O and GC-MS analysis was conducted on an Agilent 7890B GC equipped with a sniffing port (ODP 4, Gerstel, Württemberg, Germany) and an Agilent 5977B mass selective detector. The analytical columns were a DB-FFAP column (60 m × 0.25 mm i.d., 0.25 μm film thickness, Agilent, Torrance, CA).

The carrier gas was high-purity helium (>99.999 %) at a flow rate of 1.8 mL/min. The concentrated extract (1 μL) was injected into the GC injector at 250℃ in splitless mode, and the column effluent split to the mass-selective detector and sniffing port was 1:1. The transfer line temperatures and the olfactory port were set at 250℃ to prevent the condensation of high boiling point compounds. The column temperature

Table 1
Information of commercial strong-aroma baijiu from Chinese core production areas.

Sample code[a]	Region	Ingredient	ABV (%)
B1-H	Anhui	Five grains	52.0
B1-L	Anhui	Five grains	42.0
B2-H	Gansu	Five grains	52.0
B2-L	Gansu	Five grains	42.0
B3-H	Jiangsu	Five grains	52.0
B3-L	Jiangsu	Five grains	40.8
B4-H	Sichuan	Sorghum	52.0
B4-L	Sichuan	Sorghum	38.0
B5-H	Sichuan	Five grains	52.0
B5-L	Sichuan	Five grains	39.0

program was set as follows: the oven temperature was initially held at 45℃ for 2 min, raised to 150℃ at the rate of 4.5℃/min, then raised to 230℃ at the rate of 6℃/min, and held for 15 min. The temperature of the ion source was 230℃, and the ionization energy was 70 eV. The range of masses for the full-scan was used in the 35–350 amu.

The judges (three females, average age 25 years) involved in the GC-O analysis were well-trained, and they were recruited from Jiangnan University and signed the informed consent prior to their participation in the study. The study was reviewed and approved by the Jiangnan University IRB (JNU20220901IRB19). The method used in GC-O analysis was time-intensity (Van Ruth, 2001). During the GC run described above, the judges held their noses close to the sniffing port and recorded the retention time, aroma descriptors, and intensity from 0.0 to 3.0 (0 for none, 1 for weak, 2 for moderate, 3 for strong) of each odor peak. Each sample was repeated twice. The final odor-specific magnitude estimation (OSME) value for each odor peak was the average intensity of the six scoring results given by the three judges.

2.5. Identification of odor-active compounds

The compounds responsible for each odor peak were analyzed by GC−MS and comprehensive two-dimensional gas chromatography and time-of-flight mass spectrometry (GC × GC−TOFMS) analysis. GC × GC−TOFMS analysis was performed on a LECO Pegasus 4D® GC × GC−TOFMS instrument (LECO Corporation, St. Joseph, MI, USA), of which parameters had been described before (Ma et al., 2022). The following GC × GC conditions were used for the identification of the compounds that could not be detected by GC-MS. The injection mode was splitless (1.0 μL). The initial temperature of the primary oven was held at 45°C for 3 min, programmed at 4 °C/min to 150°C for 2 min and then raised at 6 °C/min to 200°C, and finally programmed at 10 °C/min to 230°C for 10 min. The temperature of the secondary oven and the temperature of the modulator were 5°C and 15°C higher than the primary oven, respectively. The modulation time was set at 4 s (0.8 s hot, 1.2 s cold pulses). Helium (99.999 %) was used as the carrier gas at a constant flow of 1.0 mL/min. The temperature of the GC injector was set to 250 °C and the temperature of the transfer line was 240 °C. The ion source was programmed at 230°C and EI voltage at 70 eV. An electron multiplier at 1500 V, an acquisition frequency of 100 spectra/s and a mass range of 35 −400 *m/z* were programmed. The Systematic LECO ChromaTOF® Workstation (version 4.44) was used for automated peak detection and spectral deconvolution. The baseline signal was above the noise, and the segmented signal-to-noise (S/N) was at 200:1 for a minimum of 2 apexing masses for peak picking. Subpeaks in the second dimension were combined if their S/N were equal to or above 6 and if they had a minimum spectral similarity above 700. The mass spectra of a selected peak with similarity scores greater than 700 were selected as a candidate peak for further identification.

Identification of odor-active compounds for each odor peak was performed by comparing the mass spectra reported in the National Institute of Standards and Technology (NIST) mass spectral library, the odor descriptions, and the Kovats retention indices (RIs) of a candidate odor-active compound with its pure standards under the same GC conditions. The Kovats retention indices (RIs) of each odor peak were calculated by injection of a reference solution of *n*-alkanes (C_6–C_{30}) under the same GC conditions, and peaks with RI differences exceeding 10 units were excluded from the peak identification.

2.6. Quantitative analysis of odor active compounds

Our methodology employs three analytical techniques which have been specifically tailored to handle the diverse and varying concentrations of compounds found in baijiu samples (Feng et al., 2025; Wang et al., 2023). The concentration of each compound was calculated from its standard calibration curve (Supplementary Table 1), which was established based on the ratio of the concentration and the peak area of the pure standard compound relative to the concentration and the peak area of the internal standard compound. Calibration standard solutions with different concentrations of the target compounds were prepared with a 1:1 dilution series in 52 % ABV solution and might be diluted to 10 % ABV according to its quantitation method. Data obtained from solutions with at least ten concentrations was subjected to regression analysis using the least squares method to construct the calibration curve. Three replicate injections of each solution were performed, and the results were averaged. The linearity, accuracy, and limit of quantification (LOQ) for each compound were evaluated. Details regarding the calibration curves and quantification methods for each compound are provided in Supplementary Table 1.

2.6.1. Gas chromatography with a flame ionization detector (GC-FID)

The quantitation of several compounds with high concentration levels (Supplementary Table 1) was achieved by direct injection using GC-FID. Compounds were separated using a CP-WAX (50 m × 0.25 mm × 0.20 μm film thickness, Agilent, Torrance, CA). A mixture of 1 mL sample and 50 μL quantitative internal standard (IS1, 2-methylbutan-2-ol, 2.33 mg/L; IS2, pentyl acetate, 2.64 mg/L) was placed in a 2 mL vial and injected into GC-FID at a split ratio of 1:20 with 1 μL each time. Helium (purity> 99.999 %) was used as the carrier gas at a flow rate of 1 mL/min and hydrogen at a flow rate of 40 mL/min. The heating-up procedure of the column oven was held at 35℃ for 5 min, ramped up to 100℃ at a rate of 4℃/min and held for 2 min, then increased to 150℃ at a rate of 8 °C/min, and finally increased to 200°C at a rate of 15 °C/min. The inlet and detector temperatures were set to 250°C. The temperature of the inlet and detector was set to 250°C. The temperature of the inlet and detector was set to 250°C.

2.6.2. Liquid-liquid microextraction followed by GC-MS (LLME-GC-MS)

Volatile organic acids, heterocycles, some alcohols, and phenols (Supplementary Table 1) were quantified by LLME coupled with GC-MS. The internal standards used in the LLME procedure were 3,3-dimethylbutanoic acid (IS3, 100 μL, 200.02 mg/L), isotopically labeled guaiacol-D3 (IS4, 40 μL, 49.99 mg/L), isotopically labeled furfural-D4 (IS5, 20 μL, 237.50 mg/L), 2-methoxy-3-(2-methylpropyl)pyrazine (IS6, 20 μL, 200.06 mg/L) and isotopically labeled hexan-1-ol-D13 (IS7, 24.6 μL, 162.70 mg/L). All baijiu samples were diluted with ultrapure water to a total alcohol content of 10 % ABV. The internal standards were added to 16 mL of each diluted baijiu sample, and the solution was then saturated with 6 g of sodium chloride solution and was oscillated and extracted with 2 mL of ethyl acetate and 2 mL of acetone for 10 min. After a period of standing, the upper organic phase was aspirated, and anhydrous sodium sulfate was added to remove the residual water; then the organic phase was concentrated under a gentle stream of nitrogen to 250 μL, and 1 μL of the concentrated extract was injected into the GC-MS. The condition of GC-MS was consistent with those described above.

2.6.3. Headspace solid-phase microextraction followed by GC-MS (HS-SPME-GC-MS)

The quantification of other compounds was achieved by using the SPME-Arrow automated headspace injection system coupled with GC-MS analysis. A 50/30 μm DVB/CAR/PDMS arrow fiber (2 cm, Supelco. Inc., USA) was used for aroma extraction. Isotopically labeled ethyl octanoate-D15 (IS8, 5 μL, 988.05 mg/L), octanal-D16 (IS9, 5 μL, 500.02 mg/L), phenylethyl acetate-D3 (IS10, 7.1 μL, 247.01 mg/L), acetophenone-D3 (IS11, 7.1 μL, 247.50 mg/L), and isopropyl disulfide (IS12, 5 μL, 12.96 mg/L) were used as internal standards. All baijiu samples were diluted with ultrapure water to a total alcohol content of 10 % (v/v), and 8 mL of diluted sample was placed into a 20 mL glass vial with a silicon septum, then added to the above internal standards and saturated with 1.5 g sodium chloride, and was equilibrated at 45°C for 5 min and extracted for 45 min under stirring at the same temperature. After extraction, the fiber was inserted into the injection port and

desorbed in non-shunt mode for 5 min. The condition of GC-MS was consistent with those described above.

To ensure clarity and consistency, the performance of our method was compared with that reported in prior studies (Feng et al., 2025; Wang et al., 2023). In terms of LOD and LOQ, our method exhibited comparable sensitivity, particularly in trace compound analysis, where SPME-GC-MS and LLME-GC-MS demonstrated advantages. GC-FID enabled robust quantification of high-abundance compounds while maintaining efficient analysis times.

2.7. Sensory analysis

The study followed the recommendations of the Declaration of Helsinki and was approved by the Jiangnan University IRB (JNU20220901IRB20). Each subject was provided an informed consent before they participated in the study, and they were paid for their participation. In all the testing sessions, samples were opened fifteen minutes before testing and were placed in a standard baijiu tasting glass with a volume of 8 mL. All the samples were labeled with a three-digit code and served at a temperature of 23°C in a randomized order.

2.7.1. Rate-all-that-apply (RATA) evaluation for different baijiu samples

All the subjects involved in this study and all the sensory analysis procedures performed in this study followed the procedures that had been published before (Xiangli et al., 2024). In summary, twenty-one subjects (11 females and 10 males) between the ages of 19 and 25 were involved, and they had good abilities to distinguish baijiu samples and different flavor attributes. Then, the subjects were asked to use the rate-all-that-apply (RATA) procedure to evaluate the samples. There were 18 attributes, namely alcohol, animal/sweaty, apple, earth, grass, honey, leather, mushroom, pineapple, qu, sorghum, sour, sweet, astringent, pungent, refreshing, soft, and bitter, involved in RATA, and the references or definitions of the attributes were published before (Xiangli et al., 2024).

2.7.2. Sensory evaluation for baijiu samples and their reconstitution samples

Key odor-active compounds were selected based on their OAVs and PLS-DA analysis. These compounds, at food-grade purity, were diluted to concentrations corresponding to their natural levels in the baijiu samples and then blended to prepare the reconstructed samples. The reconstituted baijiu samples were prepared one day prior to formal testing and stored at 23°C in nitrogen-filled sample bottles. Subsequently, all reconstructed samples were evaluated by the sensory panel using the RATA and Napping® methods.

In the Napping® experiment, twelve subjects (6 females and 6 males) were selected from the subset of the sensory panel trained for the RATA evaluation. In the formal testing procedure, all the baijiu samples and their reconstitution samples were provided to the subjects at one time. The subjects were asked to sort and place the samples on a 60 × 40 cm (A2 size) blank paper according to the similarities of their aroma attributes. If the distance between samples was closer, the aroma profile of the samples was more similar. The data of each sample was the position of the sample, which was converted to X and Y coordinates with the center of the blank paper as the origin. Then subjects were asked to select one to three distinctive aroma descriptors from the previously generated lexicon to describe each baijiu sample, and the frequency of each attribute was recorded.

2.8. Data analysis

The sensory data obtained from RATA were processed following the procedure published before (Xiangli et al., 2024). Statistical analyses of sensory data were conducted using R software (version 4.3.1, R Foundation for Statistical Computing, Vienna, Austria) involving the packages FactoMineR, Factoextra, and ComplexHeatmap. The Kruskal-Wallis test was used to analyze the significant differences of a sensory attribute between different samples. The statistical significance level in this study was set at $p < 0.05$ (significant "*": $p < 0.05$; very significant "**": $p < 0.01$; extremely significant "***": $p < 0.001$). Principal component analysis (PCA) was carried out on the RATA data to provide a global representation of the relationship between sensory attributes and different samples, and then the baijiu samples were clustered into different sensory groups according to a cluster analysis employing Euclidean distance in the clustering process. Multiple factor analysis (MFA) was carried out on the Napping® data to provide a global representation of the relationship between baijiu samples and their reconstituted samples.

A one-way analysis of variance (ANOVA) was performed using SPSS Statistics (IBM, version 22) on the concentration of each aroma compound to explore their significant differences in terms of the samples belonging to different sensory groups determined by the cluster analysis. Partial least square- discriminant analysis (PLS-DA) was carried out to discriminate between different sensory groups of strong aroma baijiu using the information from selected aroma compounds, and the method of cross-validation in PLS-DA was Jackknife (LOO).

3. Results and discussion

3.1. Sensory profile of different baijiu samples

The sensory characteristic of ten baijiu samples from five baijiu brands with two alcohol levels was described by the sensory panel using RATA with 18 attributes, and a Kruskal-Wallis test was performed on the RATA result to compare the differences in a single attribute between different samples. The result was given in Supplementary Table 2. The results showed that among all the sensory attributes, "alcohol", "apple", "mushroom", "pineapple" showed extremely significant differences ($p < 0.001$), and attributes "animal/sweaty", "soft", "honey", "sorghum" showed very significant differences ($p < 0.01$), and attributes "pungent" and "leather" showed significant differences ($p < 0.05$).

In order to clarify the characteristics and relationships between different baijiu samples in terms of their sensory attributes, all the attributes with significant differences between different samples were involved in a principal component analysis (PCA) and cluster analysis, and the result was shown in Fig. 1.

The results of PCA (Fig. 1a) showed that the first two principal components of the PCA applied to the sensory data accounted for 78.5 % of the total variance. As can be observed, baijiu samples from B1 and B3 were discriminated from baijiu samples from B2 and B5 along the PC1, and baijiu samples from B4 were discriminated from other baijiu samples along the PC2 (Fig. 1a). For the samples from B1 and B3, they had a stronger correlation with attributes "mushroom", "animal/sweaty" and "leather"; for the samples from B5, they had a stronger correlation with attributes "honey", "alcohol", "pineapple", "apple"; while for the samples from B4, they had a stronger correlation with attributes "sorghum" and "soft", which should be consistent with the fact that the samples were produced primarily with a single sorghum ingredient.

The result of the heatmap obtained from the cluster analysis (Fig. 1b) showed that if only the differences in aroma profiles were considered without considering the differences in mouthfeel, these samples could be roughly divided into three groups, where the samples from B2 were closer in overall aroma profile to the B4 samples than to the B5 samples. Among all the aroma attributes, "alcohol", "sorghum", "qu" and "pineapple" were clustered into one category, and these attributes had relatively high values in almost all the samples, indicating that these aroma attributes had some relevance in the sensory characteristics of these samples, and these attributes should be the signature aroma characteristics of a strong aroma baijiu. The attributes "honey" and "apple" were clustered into another category, and these two attributes were the distinguished attributes that could clearly differentiate the B5 samples from the other samples; meanwhile, it can be seen that "animal/sweaty",

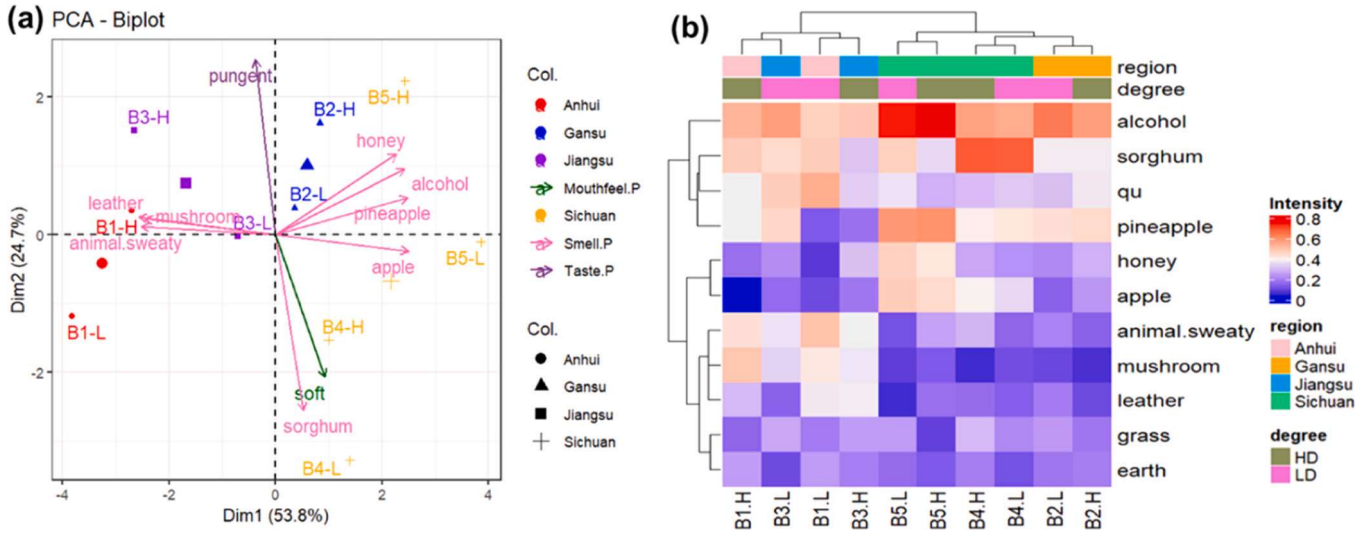


Fig. 1. Comparison of sensory characteristics of strong-aroma baijiu from different brands. (a) Principal component analysis (PCA) biplot of the selected sensory attributes of different strong-aroma baijiu samples. (b) Cluster analysis visualization and a heatmap of different strong-aroma baijiu samples considering the selected aroma attributes.

"mushroom" and "leather" were clustered into one category, and these attributes were the characteristics of the B1 and B3 samples from the other samples. In Fig. 1b, it also showed that samples from the same producer were closer together, even though they had different alcohol levels.

The typical sensory profile of strong-aroma baijiu has traditionally been associated with regional production practices and raw material composition (He et al., 2020). While previous studies have primarily focused on either intra-brand quality variations through vertical comparisons of product grades or regional distinctions through high-alcohol baijiu (Chen et al., 2024; He et al., 2020; Hong et al., 2023), our findings extend this understanding by demonstrating that commercially blended products maintain discernible geographical signatures across multiple alcohol contents. This geographical persistence in blended commercial products suggests that regional raw baijiu profiles fundamentally shape final product characteristics, even when modern blending techniques are employed to standardize alcohol content.

Although alcohol reduction typically causes significant changes in compositional profiles and flavor characteristics of alcoholic beverages (Ickes & Cadwallader, 2017; Longo et al., 2017), our PCA and Heatmap (Fig. 1) reveal minimal differentiation between samples of varying alcohol strengths within the same brand. This finding highlights the effectiveness of blending techniques employed by manufacturers to maintain product consistency. Furthermore, the transitional aroma profile of B2, which shares similarities with B4 samples, may reflect evolving trends in baijiu production, such as inter-regional raw baijiu exchanges implemented to enhance market appeal.

3.2. Identification and quantification of odor-active compounds in different strong aroma baijiu samples

In order to screen the odor-active compounds in different strong aroma baijiu samples, four samples from different aroma subgroups determined in cluster analysis were selected for GC-O, GC-MS, and GC × GC−TOFMS analysis. The extracts of the four samples exactly represented their typical aroma characteristics, indicating the reliability of the extraction procedures. The identification results of the odor-active compounds were given in Table 2. The results showed 30 esters, 13 alcohols, 10 acids, 10 aromatic compounds, 8 aldehydes, 6 pyrazines, 6 lactones, 5 phenols, 4 ketones, 4 furans, 1 sulfur-containing compound, and 1 terpene were associated with 98 odor zones (OZ), but 2 odor zones failed to be related to compounds identified by instrumental analysis, and these odor zones were carrying a toasted or floral-like odor. Specifically, the highest number of odor-active compounds was detected by GC-O in sample B5-H with 89 compounds, followed by samples B1 and B2 with 82 and 81 compounds, respectively, while the lowest number of compounds was detected in sample B3 with only 67 compounds. Of all these aroma compounds, 52 compounds were perceived in all four samples; 2-methoxy-4-methylphenol with a smoky odor and 2-methylpropyl hexanoate with a floral odor were only perceived in the B1-H sample, while hexanal and octanal with a green odor were only perceived in the B5-H sample. Among the compounds shared in all the samples, ethyl hexanoate, butanoic acid, 2,3,5-trimethylpyrazine, heptanoic acid, propyl octanoate, 2-phenylacetaldehyde, hexanoic acid, 2-methylpropanoic acid, and ethyl benzoate had relatively higher OSME values.

After identifying the main odor-active compounds in baijiu samples, the concentration of these compounds in 10 different baijiu samples was determined, except for the compound oct-1-en-3-one. The result was given in Supplementary Table 3, and it showed that the concentration of these compounds in strong aroma baijiu varied over a wide range, from ng/L to g/L. In strong aroma baijiu, the compounds with the highest average concentration, especially above 100 mg/L in most of the baijiu samples, were ethyl hexanoate, acetic acid, hexanoic acid, ethyl acetate, 1,1-diethoxyethane, and butanoic acid.

A comparison of the concentration of these compounds in baijiu samples with different alcohol levels showed that more than 70 % of the compounds, in particular nonan-2-one, (*E*)-hex-3-en-1-ol, ethyl 3-phenylpropanoate, (2*E*,6*Z*)-nona-2,6-dienal, 3-methylbutyl hexanoate, 5-ethyloxolan-2-one, and ethyl benzoate, were found in higher amounts in the baijiu samples with high-alcohol level than in the baijiu samples with low-alcohol level. However, the results showed that some fatty acid and ester compounds were found in lower amounts in some baijiu samples with high-alcohol level than in their samples with low-alcohol level, such as 2-methylpropanoic acid, 3-methylbutanoic acid, pentanoic acid, hexanoic acid, heptanoic acid, octanoic acid, nonanoic acid, ethyl 3-methylbutanoate, and ethyl heptanoate.

A comparison of the concentration of these compounds in baijiu samples from different brands showed that around 50 % of the compounds were found to be significantly different ($p < 0.05$) between the samples from different brands, which included four compounds with concentration levels more than 100 mg/L, namely, hexanoic acid, ethyl butanoate, 1,1-diethoxyethane, and butanoic acid. Although ethyl hexanoate was one of the most abundant compounds in all the strong aroma

Table 2
Identification of Odor Zones (OZs) in different strong aroma baijiu samples by GC-O, GC−MS and GC×GC−TOFMS.

OZ	RIL[a]	RI[b]	CAS	Compound	Chemical group	Odor	Identification[c]	Osme value			
								B1-H	B2-H	B3-H	B5-H
1	900	889	105–57–7	1,1-diethoxyethane	aldehydes	fruity	MS, RI, odor, Std	n.d	1.5	1.5	1.75
2	902	893	141–78–6	ethyl acetate	esters	fruity	MS, RI, odor, Std	1.75	1	1.5	2
3	961	973	97–62–1	ethyl 2-methylpropanoate	esters	fruity	MS, RI, odor, Std	1.5	1.5	1.5	1.5
4	976	994	110–62–3	pentanal	aldehydes	almond	MS, RI, odor, Std	n.d	1	n.d	2
5	1022	1014	78–92–2	butan−2-ol	alcohols	fruity	MS, RI, odor, Std	1	1	1.25	1.75
6	1025	1030	110–19–0	2-methylpropyl acetate	esters	fruity	MS, RI, odor, Std	1.5	2	1.25	1
7	1025	1035	105–54–4	ethyl butanoate	esters	fruity	MS, RI, odor, Std	2	1.5	1.75	2.25
8	1053	1057	7452–79–1	ethyl 2-methylbutanoate	esters	apple	MS, RI, odor, Std	1	1	n.d	1.5
9	1067	1061	108–64–5	ethyl 3-methylbutanoate	esters	fruity	MS, RI, odor, Std	1.75	1	2	2.75
10	1068	1065	590–86–3	3-methylbutanal	aldehydes	malt	MS, RI, odor, Std	1.75	2	1.5	2.25
11	1082	1072	123–86–4	butyl acetate	esters	fruity	MS, RI, odor, Std	1	1.25	1	1.75
12	1083	1079	66–25–1	hexanal	aldehydes	green	MS, RI, odor, Std	n.d	n.d	n.d	2
13	1108	1102	78–83–1	2-methylpropan−1-ol	alcohols	reagent, soap	MS, RI, odor, Std	1.25	1.5	2.5	1.5
14	1120	1109	71–36–3	butan−1-ol	alcohols	floral	MS, RI, odor, Std	1	1.5	2.25	1.5
15	1130	1121	6032–29–7	pentan−2-ol	alcohols	green	MS, RI, odor, Std	n.d	2	1	2.75
16	1131	1126	123–92–2	3-methylbutyl acetate	esters	banana	MS, RI, odor, Std	1.5	2	1.5	1.5
17	1159	1150	539–82–2	ethyl pentanoate	esters	apple	MS, RI, odor, Std	1	1	1.5	2
18	1208	1201	123–51–3	3-methylbutan−1-ol	alcohols	burnt	MS, RI, odor, Std	1.25	1	1	2.5
19	1240	1235	123–66–0	ethyl hexanoate	esters	pineapple, fruity	MS, RI, odor, Std	2.75	2.75	2.75	3
20	1271	1265	109–08–0	2-methylpyrazine	pyrazines	baked	RI, odor, Std	n.d	1	n.d	2
21	1277	1281	142–92–7	hexyl acetate	esters	fruity	MS, RI, odor, Std	2	1	2	n.d
22	1285	1291	4312–99–6	oct−1-en−3-one	ketones	mushroom	RI, odor, Std	2	1	2	2
23	1289	1295	106–27–4	3-methylbutyl butanoate	esters	green apple	MS, RI, odor, Std	2	n.d	1.5	2.5
24	1308	1319	108–50–9	2,6-dimethylpyrazine	pyrazines	baked	MS, RI, odor, Std	1.5	1.5	n.d	3
25	1309	1316	111–13–7	octan−2-one	ketones	fruity	MS, RI, odor, Std	1	1	n.d	n.d
26	1312	1328	659–70–1	3-methylbutyl 3-methylbutanoate	esters	fruity	MS, RI, odor, Std	2	1	1	2.25
27	1313	1331	626–77–7	propyl hexanoate	esters	fruity	MS, RI, odor, Std	n.d	1	n.d	1.25
28	1319	1335	124–13–0	octanal	aldehydes	green	MS, RI, odor, Std	n.d	n.d	n.d	1
29	1327	1342	543–49–7	heptan−2-ol	alcohols	mushroom	MS, RI, odor, Std	1.25	1	1	1.5
30	1341	1358	106–30–9	ethyl heptanoate	esters	fruity	MS, RI, odor, Std	1.75	1	2.25	2.75
31	1347	1360	105–79–3	2-methylpropyl hexanoate	esters	floral	MS, RI, odor, Std	1	n.d	n.d	n.d
32	1356	1373	111–27–3	hexan−1-ol	alcohols	floral	MS, RI, odor, Std	1.75	1.75	1.75	2.25
33	1366	1377	928–97–2	(*E*)-hex−3-en−1-ol	alcohols	green	MS, RI, odor, Std	1.5	n.d	n.d	2
34	1367	1383	821–55–6	nonan−2-one	ketones	yeast	MS, RI, odor, Std	n.d	1	n.d	1
35	1394	1389	589–98–0	octan−3-ol	alcohols	green	MS, RI, odor, Std	1	n.d	n.d	1
36	1395	1401	13925–03–6	2-ethyl−6-methylpyrazine	pyrazines	baked	MS, RI, odor, Std	n.d	1.5	n.d	1
37	1400	1412	3658–80–8	(methyltrisulfanyl)methane	sulfur-containing compounds	pickles	MS, RI, odor, Std	1.5	1	n.d	n.d
38	1408	1329	1669–44–9	(*E*)-oct−3-en−2-one	ketones	mushroom	MS, RI, odor, Std	3	1	1.5	n.d
39	1415	1436	13067–27–1	2,6-diethylpyrazine	pyrazines	nutty	MS, RI, odor, Std	1	1	n.d	1
40	1421	1420	14667–55–1	2,3,5-trimethylpyrazine	pyrazines	nutty, potato	MS, RI, odor, Std	2.5	2.5	2	2
41	1422	1428	124–19–6	nonanal	aldehydes	green	MS, RI, odor, Std	1	2	n.d	2
42	1431	1432	64–19–7	acetic acid	acids	sour	MS, RI, odor, Std	1.75	1.25	2	1.5
43	1435	1433	106–32–1	ethyl octanoate	esters	fruity	MS, RI, odor, Std	2.25	n.d	1	1.5
44	1440	1454	111–70–6	heptan−1-ol	alcohols	green	MS, RI, odor, Std	1.5	n.d	n.d	2
45	1448	1465	7779–80–8	2-methylpropyl heptanoate	esters	fruity	MS, RI, odor, Std	1.25	n.d	n.d	1.5
46	1453	1469	2198–61–0	3-methylbutyl hexanoate	esters	fruity	MS, RI, odor, Std	2	1	n.d	1.5
47	1453	1472	13529–27–6	2-(diethoxymethyl)furan	phenols	earthy	MS, RI, odor, Std	n.d	1	1	1
48	1455	1475	585–24–0	2-methylpropyl 2-hydroxypropanoate	esters	fruity	MS, RI, odor, Std	1	1	n.d	1
49	1458	1477	3391–86–4	oct−1-en−3-ol	alcohols	mushroom	MS, RI, odor, Std	2.25	2.25	1	2
50	1462	1483	1124–11–4	2,3,5,6-tetramethylpyrazine	pyrazines	coffee, nutty	MS, RI, odor, Std	1	2	1.5	1
51	1486	1489	98–01–1	furan−2-carbaldehyde	phenols	baked	MS, RI, odor, Std	2	1.5	1.75	1.25
52	1501	1495	540–07–8	pentyl hexanoate	esters	fresh fruity	MS, RI, odor, Std	2	n.d	2	2.25
53	1505	1516	112–31–2	decanal	aldehydes	fatty	MS, RI, odor, Std	2	2	1	1.75
54	1514	1521	100–52–7	benzaldehyde	aromatic alcohols	almond	MS, RI, odor, Std	1	n.d	1	1.5
55	1524	1530	624–13–5	propyl octanoate	esters	floral	MS, RI, odor, Std	2	2.75	2	2
56	1526	1536	123–29–5	ethyl nonanoate	esters	floral	MS, RI, odor, Std	n.d	2	n.d	2.5
57	1535	1539	79–09–4	propanoic acid	acids	sour	MS, RI, odor, Std	1.25	1	2	1
58	1539	1542	623–17–6	furan−2-ylmethyl acetate	phenols	green	MS, RI, odor, Std	n.d	1	n.d	1.5
59	1556	1545	108–29–2	5-methyloxolan−2-one	lactones	almond	MS, RI, odor, Std	2	1	2	1.5
60	1559	1547	111–87–5	octan−1-ol	alcohols	reagent, soap	MS, RI, odor, Std	1.5	1	1	1.25
61	1574	1588	79–31–2	2-methylpropanoic acid	acids	cheese	MS, RI, odor, Std	2	2	2.25	1.75
62	1591	1600	557–48–2	(2*E*,6*Z*)-nona−2,6-dienal	aldehydes	cucumber	MS, RI, odor, Std	1.5	2.5	2	1.5
63	1599	1520	6378–65–0	hexyl hexanoate	esters	sweet, fresh	MS, RI, odor, Std	2	n.d	1	2
64	1628	1643	107–92–6	butanoic acid	acids	cheese	MS, RI, odor, Std	3	2.5	2.25	2.5
65	1643	1659	96–48–0	oxolan−2-one	lactones	almond, coco	MS, RI, odor, Std	1.25	1	1	n.d
66	1648	1660	110–38–3	ethyl decanoate	esters	fruity, fatty	MS, RI, odor, Std	3	1	1	1.5
67	1649	1667	623–21–2	furan−2-ylmethyl butanoate	phenols	green	MS, RI, odor, Std	n.d	1.5	n.d	2
68	1650	1670	122–78–1	2-phenylacetaldehyde	aromatic alcohols	fruity	MS, RI, odor, Std	2.5	2.5	1	2.75

(*continued on next page*)

Table 2 (*continued*)

OZ	RIL[a]	RI[b]	CAS	Compound	Chemical group	Odor	Identification[c]	Osme value			
								B1-H	B2-H	B3-H	B5-H
69	1652	1671	93–89–0	ethyl benzoate	aromatic alcohols	floral	MS, RI, odor, Std	2	1.5	1.5	3
70	1655	1673	98–86–2	1-phenylethanone	aromatic alcohols	floral	MS, RI, odor, Std	1	1	1.5	2
71	1667	1680	143–08–8	nonan−1-ol	alcohols	green, earthy	MS, RI, odor, Std	1.5	2	1	1.5
72	1662	1682	503–74–2	3-methylbutanoic acid	acids	cheese	MS, RI, odor, Std	1	1	1. 5	1.25
73	1694	1690	123–25–1	diethyl butanedioate	esters	fruity	MS, RI, odor, Std	3	1	1	1.5
74	1703	1721	695–06–7	5-ethyloxolan−2-one	lactones	nutty	MS, RI, odor, Std	1.5	1	2	1
75	1733	1749	109–52–4	pentanoic acid	acids	cheese, sour	MS, RI, odor, Std	2	1.5	1.5	2
76	1755	1764	627–90–7	ethyl undecanoate	esters	sweet	MS, RI, odor, Std	2	1	n.d	2
77	1785	1799	101–97–3	ethyl 2-phenylacetate	aromatic alcohols	rose	MS, RI, odor, Std	1.5	2	1.5	2.5
78	1796	1806	105–21–5	5-propyloxolan−2-one	lactones	floral, nutty	MS, RI, odor, Std	1.5	1	n.d	n.d
79	1827	1840	23726–93–4	β-damascenone	terpenes	floral	MS, RI, odor, Std	n.d	1	n.d	2
80	1835	1842	103–45–7	2-phenylethyl acetate	aromatic alcohols	honey, floral	MS, RI, odor, Std	1	n.d	n.d	1
81	1857	1860	142–62–1	hexanoic acid	acids	sour	MS, RI, odor, Std	2	2	2.5	2
82	1860	1871	106–33–2	ethyl dodecanoate	esters	floral	MS, RI, odor, Std	1.5	2	1.75	2
83	1895	1900	100–51–6	phenylmethanol	aromatic alcohols	sweet	MS, RI, odor, Std	n.d	2	n.d	1.5
84	1900	1919	2021–28–5	ethyl 3-phenylpropanoate	aromatic alcohols	floral, honey	MS, RI, odor, Std	n.d	2	2	2
85	1911	1931	104–50–7	5-butyloxolan−2-one	lactones	sweet, baked	MS, RI, odor, Std	1.5	1	1.5	1.25
86	1931	1939	60–12–8	2-phenylethanol	aromatic alcohols	honey, floral	MS, RI, odor, Std	2.75	2	1	2
87	1956	1941	93–51–6	2-methoxy−4-methylphenol	phenols	smoky	MS, RI, odor, Std	2	n.d	n.d	n.d
88	1960	1978	111–14–8	heptanoic acid	acids	cheese	MS, RI, odor, Std	2.5	1	3	2.5
89	2018	2029	104–61–0	5-pentyloxolan−2-one	lactones	fruity	MS, RI, odor, Std	1.25	1	1.25	1
90	2048	2054	2785–89–9	4-ethyl−2-methoxyphenol	phenols	smoky	MS, RI, odor, Std	2	1	1.5	n.d
91	2053	2069	124–06–1	ethyl tetradecanoate	esters	hay	MS, RI, odor, Std	2	n.d	1.5	2.25
92	2070	2091	124–07–2	octanoic acid	acids	sour	MS, RI, odor, Std	2	1.5	2	2
93	2089	2097	106–44–5	4-methylphenol	phenols	smoky, leather	MS, RI, odor, Std	1	2	2	1
94	2175	2192	112–05–0	nonanoic acid	acids	sour	MS, RI, odor, Std	2.25	1	1	1.75
95	2189	2199	7786–61–0	4-ethenyl−2-methoxyphenol	phenols	smoky	MS, RI, odor, Std	1	n.d	1.25	1
96	2195	2210	123–07–9	4-ethylphenol	phenols	leather	MS, RI, odor, Std	1.5	2	2	1
97	2240	2257	628–97–7	ethyl hexadecanoate	esters	cream	MS, RI, odor, Std	2	n.d	n.d	1.75
98	2571	2582	121–33–5	4-hydroxy−3-methoxybenzaldehyde	aromatic alcohols	vanilla	MS, RI, odor, Std	1	1	1	1
		1470	-	unknown	-	roast meat	-	n.d	n.d	n.d	2
		1490	-	unknown	-	floral	-	n.d	n.d	n.d	1

[a] The retention index (RIL) is derived from DB-FFAP in the literature.
[b] The actual retention index (RI) is calculated by Masshunter (Agilent, America).
[c] Identification of odor-active compounds for each odor peak was performed by comparing the mass spectra reported in the National Institute of Standards and Technology (NIST) mass spectral library (Abbreviated as MS), the odor descriptions (Abbreviated as odor), and the Kovats retention indices (RIs) of a candidate odor-active compound with its pure standards under the same GC conditions (Abbreviated as Std).

baijiu samples, its concentration did not differ significantly between the samples from different brands.

This study's chemical characterization of strong-aroma baijiu elucidates the relationships between aroma compounds, alcohol content, and brand-specific profiles, thereby complementing existing research in the field. Consistent with previous findings (Chen et al., 2024), our results confirm the predominance of esters and related compounds, particularly ethyl hexanoate, hexanoic acid, ethyl acetate, and butanoic acid. However, our analysis further demonstrates significant concentration variations of these compounds across different alcohol concentrations and brands. Our results also align with another study in emphasizing the technical challenges of detecting minor polar compounds, such as pyrazines and furans (He et al., 2021). These compounds, though present in lower concentrations, contribute distinct flavor notes, such as "toasted" or "smoky" odors, which were detected in our sensory evaluation but some of them might remain unassigned in instrumental analysis.

3.3. Correlation between odor-active compounds and aroma cluster analysis groups of baijiu samples

Based on the results of the cluster analysis of the RATA data, the 10 samples were roughly divided into three major groups, of which group I was the sample from brands B1 and B3, group II was the sample from brands B2 and B4, and group III was the sample from brand B5. In order to determine which compounds were more important in the discrimination of the samples belonging to the different cluster groups, one-way ANOVA and PLS-DA were performed. One-way ANOVA was first conducted to select the odor-active compounds that were significantly different ($p < 0.05$) between the sample groups, and the results were given in Table 3. It showed that 38 odor-active compounds were significantly different between different sample groups, and among all these compounds, ethyl 2-methylpropanoate, hexanal, ethyl pentanoate, ethyl heptanoate, hexan-1-ol, 2-ethyl-6-methylpyrazine, nonanal, heptan-1-ol, furan-2-ylmethyl butanoate, hexanoic acid, and octanoic acid showed extremely significant differences ($p < 0.001$).

In order to determine which compounds were more important in the discrimination of the different sample groups, PLS-DA was carried out to discriminate the selected odor-active compounds related to the cluster groups, and the result map was given in Fig. 2. The result (Fig. 2a) showed that PLS-DA score plots well separated group III from other groups, but the PLS-DA model didn't well separate group I and group II. The values of R^2X, R^2Y for the PLS-DA model, which indicates the model interpretation rate, were 0.534 and 0.713, and Q^2 for the PLS-DA model, which represents the model predictive ability, was 0.531. Fig. 2b and Fig. 2c provided the most important odor-active compounds in discriminating different cluster groups based on components 1 and 2 of the PLS-DA model. Fig. 2b and Fig. 2c showed that 26 compounds, including nonanal, 2-ethyl-6-methylpyrazine, hexan-1-ol, hexanoic acid, octanoic acid, ethyl pentanoate, etc., were the most important in discriminating samples from different cluster groups.

Cluster analysis and subsequent PLS-DA modeling revealed distinct aroma compound distribution patterns among the three sample groups. Group III (B5) exhibited a unique chemical profile that enabled clear differentiation from groups I (B1, B3) and II (B2, B4). Eleven odor-active compounds, including ethyl pentanoate, nonanal, and hexanoic acid, demonstrated particularly high discriminatory significance ($p < 0.001$),

Table 3
Odor-activity values (OAVs) of compounds identified by GC-O and GC-MS in different strong aroma baijiu samples.

Odor Zone	Compounds[α]	Threshold (μg/L)	Cluster group 1				Cluster group 2				Cluster group 3		p value[β]
			B1-H	B1-L	B3-H	B3-L	B2-H	B2-L	B4-H	B4-L	B5-H	B5-L	
19	ethyl hexanoate[#]	55.3[a]	29229	23061	28017	22810	28456	22909	27266	22513	30552	25546	
7	ethyl butanoate[#]	81.5[a]	3299	2679	3069	310	2990	264	2796	2403	5348	3030	**
10	3-methylbutanal[#]	16.5[a]	1211	525	1077	332	1177	366	688	135	1159	565	
81	hexanoic acid[#]	2520[a]	452	467	449	616	476	559	366	268	301	283	***
9	ethyl 3-methylbutanoate[#]	6.89[a]	376	847	261	871	449	728	133	236	494	313	
17	ethyl pentanoate[#]	26.8[a]	399	312	645	471	337	221	242	370	644	543	***
43	ethyl octanoate[#]	12.9[a]	692	160	497	171	342	116	177	217	412	236	
62	(2*E*,6*Z*)-nona−2,6-dienal[#]	0.64[b]	196	43	408	7.9	592	7.3	216	86	463	199	
37	(methyltrisulfanyl)methane[#]	0.36[a]	186	166	347	< 1	368	< 1	43	< 1	205	204	
3	ethyl 2-methylpropanoate[#]	57.5[a]	139	65	169	81	131	74	157	55	262	143	***
64	butanoic acid[#]	965[a]	153	135	91	128	156	105	157	127	100	92	**
75	pentanoic acid	389[a]	61	90	60	93	64	87	48	102	84	55	
1	1,1-diethoxyethane[#]	2090[a]	54	56	91	72	60	59	43	58	61	76	*
12	hexanal[#]	25.5[a]	32	7.2	27	10	31	10	53	6.8	121	33	***
8	ethyl 2-methylbutanoate[#]	18.0[a]	59	10	29	16	51	12	40	6.1	65	17	
14	butan−1-ol	2733[a]	40	17	35	4.7	50	4.9	29	3.7	64	11	
2	ethyl acetate[#]	32552[a]	29	16	28	16	27	20	28	24	32	27	*
49	oct−1-en−3-ol	6.12[a]	32	22	2.5	15	42	21	20	15	17	15	
79	β-damascenone	0.12[c]	< 1	31	< 1	29	37	< 1	< 1	19	42	34	**
16	3-methylbutyl acetate[#]	93.9[a]	41	19	13	10	18	16	5	2.8	15	12	*
84	ethyl 3-phenylpropanoate	125[a]	21	1.1	16	< 1	26	< 1	10	4.9	47	< 1	
63	hexyl hexanoate[#]	1891[d]	17	17	6.9	17	4	26	3.4	14	16	6.2	
72	3-methylbutanoic acid[#]	1050[a]	12	15	6.5	12	9.7	13	4.5	14	7.3	6.3	*
32	hexan−1-ol[#]	5370[d]	4	9.5	9.6	9.1	9.9	9.5	9.9	8.2	17	10	***
77	ethyl 2-phenylacetate[#]	407[a]	12	3.8	10	4.3	15	4.2	4.9	1.1	36	6.1	**
53	decanal	70.8[e]	29	< 1	1.8	< 1	< 1	< 1	< 1	< 1	14	1.2	
42	acetic acid[#]	160000 [f]	8.6	1.9	9	1.8	8.5	2	4.4	1.8	8.5	< 1	
82	ethyl dodecanoate[#]	400 [g]	3.6	10	< 1	4.7	< 1	13	< 1	13	< 1	< 1	
46	3-methylbutyl hexanoate	1400[e]	4.7	5.2	1.5	< 1	1.2	< 1	< 1	2.3	3.5	3.6	**
68	2-phenylacetaldehyde	262[d]	7.9	< 1	3.6	< 1	7.1	< 1	3.1	< 1	12	2.2	
61	2-methylpropanoic acid	1580 [f]	7.4	3.7	3.2	3.3	4	4.5	1.7	3.5	3	2	*
92	octanoic acid	2700[a]	8.4	5.9	4.2	4.6	2.1	5.9	< 1	2.8	< 1	< 1	***
93	4-methylphenol	167[a]	8.2	1.1	4.2	1.1	4.2	1.2	2.4	< 1	5.8	< 1	
56	ethyl nonanoate	3151[a]	3.7	< 1	2.5	1.8	4.1	< 1	1.8	< 1	11	1.4	**
11	butyl acetate	1800[h]	3.3	2.1	1.7	1	3.5	1.8	4	1.1	4.3	2.1	
66	ethyl decanoate	1120[a]	6.1	3.8	< 1	2.7	< 1	1.3	< 1	4.6	1.9	< 1	*
30	ethyl heptanoate[#]	13153[a]	1.4	1.8	< 1	1.4	< 1	1.3	< 1	1.8	2.8	1.6	***
13	2-methylpropan−1-ol[#]	28300 [f]	3.2	< 1	1.7	< 1	1.8	< 1	< 1	< 1	3.1	1.6	**
36	2-ethyl−6-methylpyrazine	40.0[d]	1.8	1.8	1.5	1.5	1.4	1.4	1.4	1.4	< 1	< 1	***
88	heptanoic acid	13281[a]	1	3	< 1	2	< 1	2.3	< 1	2.1	1	< 1	*
39	2,6-diethylpyrazine	6.00[d]	2.6	< 1	2.5	< 1	2	< 1	2.3	< 1	3.4	< 1	
23	3-methylbutyl butanoate	915 [f]	2.3	< 1	1.9	< 1	1.5	< 1	2.4	< 1	3.2	< 1	
27	propyl hexanoate	12784[a]	1.5	< 1	1.5	< 1	1.5	< 1	1.9	< 1	2.3	< 1	
97	ethyl hexadecanoate	2000[a]	2.8	< 1	1.2	< 1	1.6	< 1	< 1	< 1	2.1	< 1	
6	2-methylpropyl acetate	922[a]	1.5	< 1	1.1	< 1	2.2	< 1	2	< 1	1.1	< 1	
28	octanal[#]	39.6[a]	1.4	< 1	1.2	< 1	< 1	< 1	1.8	< 1	3.7	< 1	
89	5-pentyloxolan−2-one[#]	90.7[a]	2.1	< 1	1.8	< 1	1.1	< 1	1	< 1	1.8	< 1	
60	octan−1-ol	1100[d]	1.3	1.6	< 1	< 1	< 1	1.1	< 1	2.3	< 1	< 1	
38	(*E*)-oct−3-en−2-one	48.2[b]	3.7	< 1	1.9	< 1	2.8	< 1	< 1	< 1	< 1	< 1	
76	ethyl undecanoate	1000 [l]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	
55	propyl octanoate	662[a]	1.4	< 1	1.5	< 1	1.2	< 1	1.4	< 1	1.6	< 1	
29	heptan−2-ol[#]	1431[a]	1.1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	1.1	< 1	**
5	butan−2-ol	50000 [m]	1	< 1	< 1	< 1	< 1	< 1	1	< 1	1.6	< 1	
51	furan−2-carbaldehyde	44000[a]	1.3	< 1	1.1	< 1	< 1	< 1	1.1	< 1	< 1	< 1	
33	(*E*)-hex−3-en−1-ol	400 [n]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	1.2	< 1	
47	2-(diethoxymethyl)furan	6172 [f]	1.6	< 1	2.6	< 1	2.4	< 1	< 1	< 1	2.4	< 1	
18	3-methylbutan−1-ol	179191[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	**
57	propanoic acid	18200[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	*
96	4-ethylphenol	123[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	1.9	< 1	< 1	
21	hexyl acetate	5560[d]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	*
90	4-ethyl−2-methoxyphenol	123[a]	1.1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	**
94	nonanoic acid	3559[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	
34	nonan−2-one	393[d]	< 1	< 1	< 1	< 1	1.2	< 1	< 1	< 1	< 1	< 1	
80	2-phenylethyl acetate	909[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	
35	octan−3-ol	483[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	
71	nonan−1-ol	806[d]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	
48	2-methylpropyl 2-hydroxypropanoate	4904[b]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	
70	1-phenylethanone	256[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	
65	oxolan−2-one	20000 [m]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	
26	3-methylbutyl 3-methylbutanoate	134[b]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	
24	2,6-dimethylpyrazine	791[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	

(*continued on next page*)

Table 3 (*continued*)

Odor Zone	Compounds[α]	Threshold (μg/L)	Cluster group 1				Cluster group 2				Cluster group 3		p value[β]
			B1-H	B1-L	B3-H	B3-L	B2-H	B2-L	B4-H	B4-L	B5-H	B5-L	
87	2-methoxy−4-methylphenol	315[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	
40	2,3,5-trimethylpyrazine	730[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	
54	benzaldehyde	4203[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	
15	pentan−2-ol	194313[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	*
4	pentanal	725[d]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	**
31	2-methylpropyl hexanoate	5350[d]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	
41	nonanal	122[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	***
44	heptan−1-ol	26600[i]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	***
98	4-hydroxy−3-methoxybenzaldehyde	439[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	**
52	pentyl hexanoate	13802[d]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	**
95	4-ethenyl−2-methoxyphenol	209[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	
86	2-phenylethanol	28900[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	
69	ethyl benzoate	407[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	**
25	octan−2-one	250 [l]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	
73	diethyl butanedioate	353193[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	
20	2-methylpyrazine	60000[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	
67	furan−2-ylmethyl butanoate	4900[d]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	***
74	5-ethyloxolan−2-one	12500[d]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	
78	5-propyloxolan−2-one	1000[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	
85	5-butyloxolan−2-one	2816[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	
59	5-methyloxolan−2-one	2200[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	
58	furan−2-ylmethyl acetate	26050 [p]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	
91	ethyl tetradecanoate	33551[b]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	*
83	phenylmethanol	40927[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	*
50	2,3,5,6-tetramethylpyrazine	80073[a]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	*
45	2-methylpropyl heptanoate	26264[d]	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	

Note: [α]Compounds marked with "#" were selected in the aroma reconstitution experiment.
[β]Odor threshold previously reported in different literatures:
γ One-way ANOVA was performed on concentration of each compound to check the significantly different ($p < 0.05$, "*"; $p < 0.01$, "**"; $p < 0.001$, "***") between the groups determined in aroma cluster analysis.
[a] from reference (Fan & Xu, 2011);
[b] measured in a 46 % water/ethanol;
[c] from reference (Pineau et al., 2007);
[d] from reference (Feng et al., 2025);
[e] from reference (Lin et al., 2024);
[f] from reference (Gao et al., 2014);
[g] from reference (Chen et al., 2022);
[h] from reference (Zhou, 2019);
[i] from reference (Fan et al., 2015);
[l] from reference (Tan & Siebert, 2004);
[m] from reference (Peinado et al., 2004);
[n] from reference (Zhao et al., 2019);
[p] from reference (Dong, 2020).

suggesting their potential as chemical markers for group III identification. The partial overlap between groups I and II in the PLS-DA score plot (Fig. 2a) indicates either shared chemical characteristics or insufficient variability in key discriminators between these clusters. This observation correlates with the model's moderate predictive capacity (Q^2=0.531), which, while statistically acceptable, underscores the challenges in establishing robust correlations between chemical profiles and sensory groupings for strong-aroma baijiu.

While the R^2Y value in this study demonstrates reasonable explanatory power for aroma perception variability, the limited separation between groups I and II suggests the potential influence of additional factors on sensory outcomes. These may include perceptual interactions among compounds or contributions from unmeasured volatile components. This finding contrasts with previous successful applications of PLS-DA for geographical origin authentication, where chemical markers showed more direct correlations with region-specific production practices (Song et al., 2020). The current results align with emerging understanding that sensory perception involves multidimensional integration of odorants, which often exhibits non-linear relationships with individual compound concentrations (Ma et al., 2025). Our identification of 26 influential compounds in components 1 and 2 (Fig. 2b, c) establishes a prioritized framework for future mechanistic investigations into baijiu flavor perception.

3.4. Aroma reconstitution experiments to verify the contribution of important odor-active compounds

The odor activity values (OAVs) of each compound were calculated to determine the contribution of each compound in different baijiu samples. The result showed that the OAVs of 59 compounds were greater than 1 in at least 1 sample, and 29 of them were greater than 10. Among all the samples, sample B1-H had the highest number of compounds with OAVs greater than 1, while sample B5-H had the highest number of compounds with OAVs greater than 10. Ethyl hexanoate, ethyl butanoate, 3-methylbutanal, ethyl pentanoate, ethyl 3-methylbutanoate, hexanoic acid, and ethyl octanoate were the most important odorants in strong aroma baijiu with the OAVs above 100 in almost all the samples, and the OAVs of (methyltrisulfanyl)methane, (2*E*,6*Z*)-nona-2,6-dienal, ethyl 2-methylpropanoate, and butanoic acid were found to be greater than 100 in some samples but less than 100 in others.

After obtaining the OAVs for each compound, an aroma reconstitution experiment was carried out to confirm the contribution of the most important odor-active compounds to the odor profile of a strong-aroma baijiu and to verify if these selected compounds could mimic the aroma

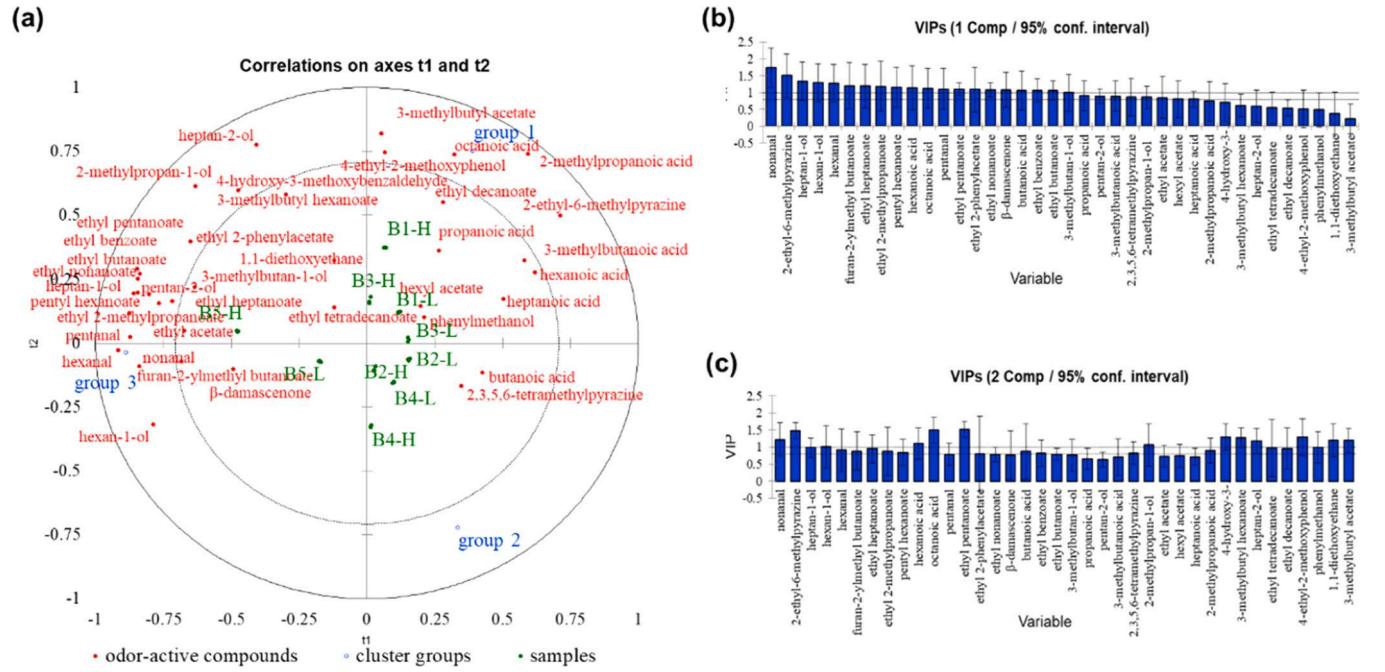


Fig. 2. PLS-DA result which explained the selected odor-active compounds related to the cluster groups. (a) PLS-DA score plots of the baijiu samples and the selected odor-active compounds using the first two components as axes. (b) Loading plot applied to the odor-active compounds to discriminate baijiu samples from different cluster groups based on the components 1 of the PLS-DA model. (cd) Loading plot applied to the odor-active compounds to discriminate baijiu samples from different cluster groups based on the components 2 of the PLS-DA model.

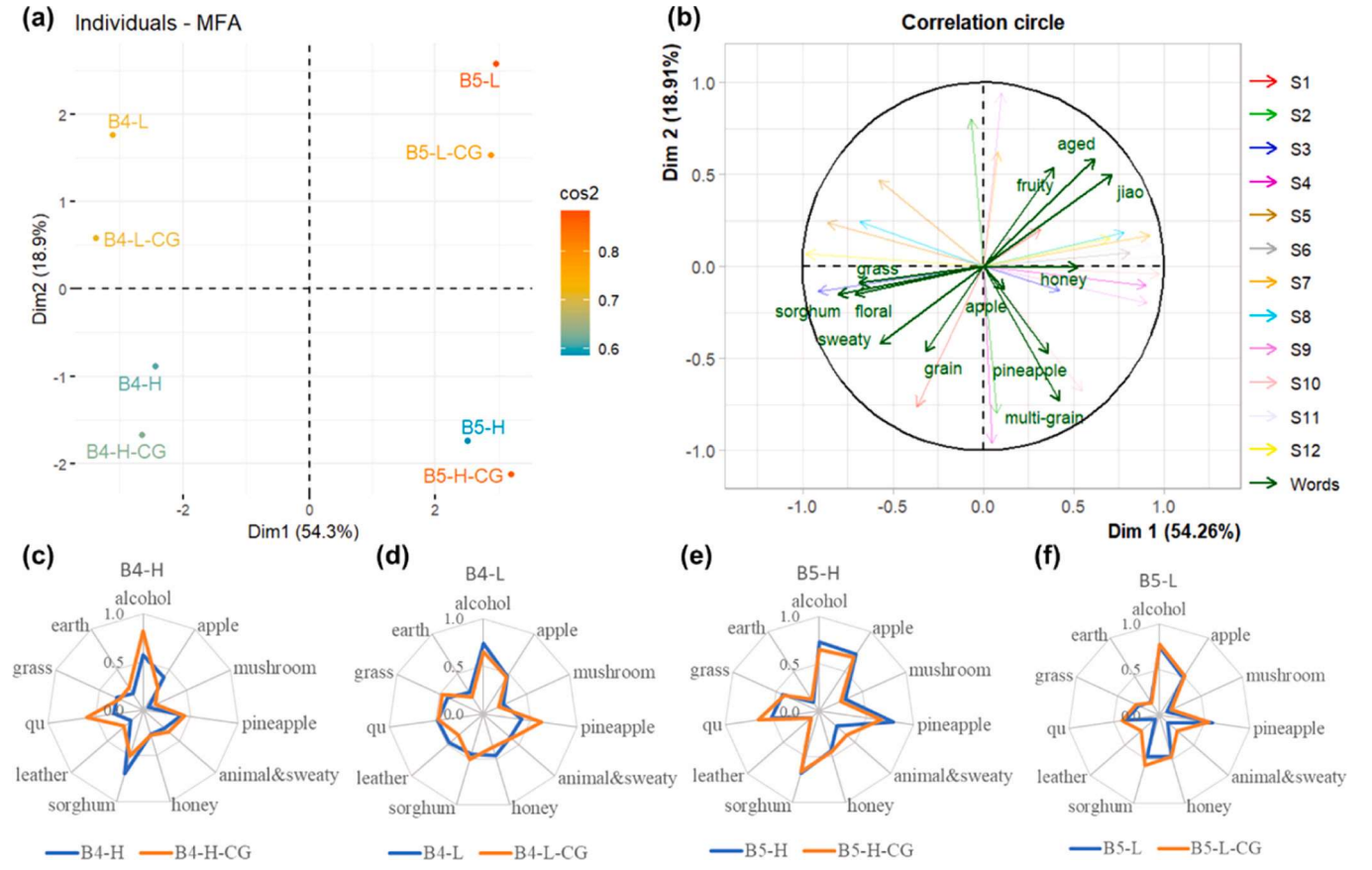


Fig. 3. Comparison of sensory characteristics of strong aroma baijiu samples and their reconstitution samples. (a) individual factor map of multiple factor analysis (MFA) of Napping®; (b) correlation circle of multiple factor analysis (MFA) of Napping®; (c) the odor profile of B4-H sample and its reconstitution sample evaluated by RATA (B4-H-CG); (d) the odor profile of B4-L sample and its reconstitution sample evaluated by RATA (B4-L-CG); (c) the odor profile of B5-H sample and its reconstitution sample evaluated by RATA (B5-H-CG); (c) the odor profile of B5-L sample and its reconstitution sample evaluated by RATA (B5-L-CG).

profiles of samples from different sensory clustering groups. Twenty-seven odor-active compounds marked in Table 3 were selected to reconstruct four samples from the Sichuan production area based on their OAVs and their significant differences among different sensory clustering groups. A napping® evaluation was conducted on the four baijiu samples and their reconstitution samples, and the sensory results were shown in Fig. 3. As shown in Fig. 3a, the overall characteristics of the aromas of the four different samples and their reconstitution samples could be clearly distinguished, and the overall aroma perception of each sample over its reconstitution sample was basically close. Fig. 3b showed that although the subjects used some different descriptors in the napping experiment, the results were like those in the first RATA experiment.

In order to clarify the performance of these aroma compounds in mimicking the different aroma attributes of the four samples, the aroma characteristics of the four baijiu samples and their reconstitution samples were described again by using RATA, and the results were given in Fig. 3c-f. The results proved that the aroma profile of each reconstitution sample was approximately close to their original sample, especially for the sample with high alcohol, but still had some intensity difference with its original sample on some attributes such as "qu", "pineapple", "leather", and "animal".

3.5. General discussion

The aroma characteristic of strong-aroma baijiu has been studied before, and these studies focused on different grades of samples from one brand or raw strong-aroma baijiu from different regions with higher alcohol content (Chen et al., 2024; He et al., 2020; Hong et al., 2023). In these studies, it showed that within a brand, higher quality strong-aroma baijiu tended to have stronger aroma perception in terms of "fruity", "flower", "aged" and "roast" attributes and had weaker aroma perception in terms of "mud pit" and "acids" (Chen et al., 2024; Hong et al., 2023); across brands, the brand of strong aroma baijiu from the upper reaches of the Yangtze River (e.g., Sichuan province) had significant differences in sensory characteristics from the brand of strong aroma baijiu from the lower reaches of the Yangtze River (e.g., Jiangsu province) (He et al., 2020; Xiangli et al., 2024). In our study, we compared the commercial samples from two of the most representative brands from these two regions, not only the samples with higher alcohol levels, but also the samples with lower alcohol levels, and the result showed (Fig. 1b) that although the commercial baijiu was obtained by combining different raw baijiu, the aroma style of the commercial strong-aroma baijiu was still characterized by regional differences.

The aroma compounds in different strong-aroma baijiu also have been studied before (Chen et al., 2024; He et al., 2021; Hong et al., 2023; Zhao et al., 2018), and compared to their studies, the present study is not more innovative in terms of the means of detection and the types of compounds detected. Notably, comparative analysis of compounds and their concentrations across different alcohol concentrations within the same brand revealed remarkably similar sensory aroma profiles. This consistency persisted despite substantially higher concentrations of most volatile compounds in high-alcohol samples compared to their low-alcohol counterparts. This result proved that the relative proportions between compounds within a certain concentration range should be more important than the absolute concentration of a compound in baijiu aroma perception, especially when the change of its concentration cannot break through the aroma buffering effect of an alcoholic beverage (Ferreira et al., 2022; Ma et al., 2025).

To investigate the relationship between compound concentrations and perceived aroma differences, we performed quantitative analysis of key aroma compounds across samples exhibiting distinct sensory profiles. These data were subsequently employed to develop a PLS-DA model capable of predicting overall aroma perception differences based on variations in target compound concentrations. The results showed that component 1 in the PLS-DA model predicted better for group 3, but the overall performance of this predictive model was merely acceptable. Previous studies had established a PLS-DA model to predict the classification of strong-aroma baijiu according to geographical origin, and the model gave a predictive capability value of 0.982 (Song et al., 2020), but in terms of articles published so far, there is no model that has yet achieved an acceptable result in predicting the sensory properties of strong-aroma baijiu from its chemical profiles. In our study, PLS regression was employed to examine relationships between sensory characteristics and chemical compounds, however, the predictive performance for certain aroma attributes proved limited. This likely reflects both the inherent complexity of aroma perception and constraints imposed by our sample size. Consequently, these results have not been presented in detail. Our findings underscore the necessity for more comprehensive datasets incorporating diverse brands, coupled with advanced multivariate analytical approaches, to better elucidate the nonlinear relationships between chemical composition and sensory properties in complex spirits such as baijiu.

While the current models provide valuable preliminary insights, substantial challenges remain in reconciling instrumental measurements with sensory evaluations (Taylor, 2023). Future research should focus on three critical aspects: (1) expanding sample sets to encompass greater product variability, (2) employing machine learning approaches to better capture complex nonlinear relationships, and (3) systematically accounting for perceptual interactions among odorants. Such multidimensional investigations will be crucial for advancing our understanding of baijiu flavor chemistry.

4. Conclusions

In summary, this study represents a significant advancement in the understanding of strong-aroma baijiu' aroma characteristics by combining multi-analytical techniques and sensory evaluations. Rate-all-that-apply (RATA) revealed extremely significant differences in attributes "alcohol", "apple", "mushroom" and "pineapple" among samples from different brands, and the ten samples from different brands could be clustered into three groups based on their global aroma profiles. A total of 97 compounds were identified and quantitated by GC-O combined with GC-FID, GC-MS, and GC×GC-TOFMS analysis, and 26 compounds, including nonanal, 2-ethyl-6-methylpyrazine, hexan-1-ol, hexanoic acid, octanoic acid, ethyl pentanoate, etc., were the most important in discriminating samples from different groups based on the PLS-DA result. Aroma reconstruction experiments based on OAVs confirmed the contribution of 27 compounds, including ethyl hexanoate, ethyl butanoate, 3-methylbutanal, ethyl pentanoate, ethyl 3-methylbutanoate, hexanoic acid, and ethyl octanoate, etc., to the aroma profiles of strong-aroma baijiu, particularly for high-alcohol samples. The observation obtained in this study should provide a foundation for future research in aromatic modulation and quality control in the baijiu industry.

Ethical statements

Ethical approval for the involvement of human subjects in this study was granted by Jiangnan University Research Ethics Committee (Approval number: JNU20220901IRB19 and JNU20220901IRB20), and all procedures strictly adhered to the Declaration of Helsinki principles. The informed consent form was provided to all participants prior to their involvement in the study, and they were free to withdraw at any time without penalty. In our study, no vulnerable populations (children, institutionalized individuals, or persons with diminished capacity) were involved, and no personally identifiable information was collected or disclosed.

Funding

This work was supported by the National Natural Science Foundation

of China for Young Scholars [32202218], the Fundamental Research Funds for the Central Universities [JUSRP123038].

CRediT authorship contribution statement

Yue MA: Writing – original draft, Visualization, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Ran XIANGLI:** Writing – original draft, Validation, Methodology, Investigation, Formal analysis. **Dong ZHAO:** Writing – review & editing, Supervision, Resources, Conceptualization. **Jia ZHENG:** Resources, Methodology, Conceptualization. **Jian SU:** Writing – review & editing, Resources, Methodology. **Yan XU:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors thank all the subjects who participated in the sensory experiments.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jfca.2025.108076.

Data availability

Data will be made available on request.

References

Chen, E.B., Yang, F., Ma, Z.Y., Geng, X.J., Li, Q., Zheng, F.P., Sun, J.Y., Sun, B.G., 2024. Analysis of aroma compounds in different grades of Luzhou Laojiao strong-aroma Baijiu by molecular sensory science. J. Food Compos. Anal. 135, 106607. https://doi.org/10.1016/j.jfca.2024.106607.

Chen, P.P., Liu, Y., Wu, J.H., Yu, B., Zhao, H., Huang, M.Q., Zheng, F.P., 2022. Sensory-directed decoding of key aroma compounds from Jiugui-series Baijiu, the representative of Fuyu-flavor-type Baijiu (FFTB). J. Food Compos. Anal. 114, 104799. https://doi.org/10.1016/j.jfca.2022.104799.

Dong, W., 2020. The studies of the key odorants causing. JiaoAroma their Form. Pathw. StrongAroma Types baijiu. [Dr. Diss. South China Univ. Technol. https://doi.org/10.27151/d.cnki.ghnlu.2020.000212.

Fan, H.Y., Fan, W.L., Xu, Y., 2015. Characterization of key odorants in Chinese chixiang aroma-type liquor by gas chromatography–olfactometry, quantitative measurements, aroma recombination, and omission studies. J. Agric. Food Chem. 63 (14), 3660–3668. https://doi.org/10.1021/jf506238f6238f.

Fan, W.L., Qian, M.C., 2005. Headspace solid phase microextraction and gas chromatography– olfactometry dilution analysis of young and aged Chinese "Yanghe Daqu" liquors. J. Agric. Food Chem. 53 (20), 7931–7938. https://doi.org/10.1021/jf051011k.

Fan, W.L., Qian, M.C., 2006. Characterization of aroma compounds of Chinese "Wuliangye" and "Jiannanchun" liquors by aroma extract dilution analysis. J. Agric. Food Chem. 54 (7), 2695–2704. https://doi.org/10.1021/jf052635t.

Fan, W.L., Xu, Y., 2011. Determination of odor thresholds of volatile aroma compounds in baijiu by a forced-choice ascending concentration series method of limits. Liquor Mak. 38 (4), 80–84.

Fang, C., Xu, Y., 2023. Classification of Chinese Baijiu. In: Xu, Y. (Ed.), Science and engineering of Chinese liquor (Baijiu): Microbiology, chemistry and process technology. Springer, Singapore, pp. 43–63.

Feng, Y.F., Zhang, Z.L., Huang, X.D., Tang, L., Guo, T.H., Zhang, J., Chen, S., Xu, Y., 2025. Molecular and sensory characterization of Chinese Baijiu aged in Jiuhai container and contribution of volatile odorous terpenes to Jiuhai-aroma. Food Chem. 470, 142687. https://doi.org/10.1016/j.foodchem.2024.142687.

Ferreira, V., de la Fuente, A., Sáenz-Navajas, M.P., 2022. Wine aroma vectors and sensory attributes. In: Reynolds, A.G. (Ed.), Managing wine quality: Viticulture and wine quality, Vol. 1. Woodhead Publishing, pp. 3–39. https://doi.org/10.1016/B978-0-08-102067-8.00008-7.

Gao, W.J., Fan, W.L., Xu, Y., 2014. Characterization of the key odorants in light aroma type Chinese liquor by gas chromatography–olfactometry, quantitative measurements, aroma recombination, and omission studies. J. Agric. Food Chem. 62 (25), 5796–5804. https://doi.org/10.1021/jf501214c.

Granvogl, M., Schieberle, P., 2022. The sensomics approach: A useful tool to unravel the genuine aroma blueprint of foods and aroma changes during food processing. In: Chiara Emilia Irma Cordero (Ed.), Comprehensive analytical chemistry, 96. Elsevier, pp. 41–68. https://doi.org/10.1016/bs.coac.2021.10.002.

He, Y.X., Liu, Z.P., Qian, M.C., Yu, X.W., Xu, Y., Chen, S., 2020. Unraveling the chemosensory characteristics of strong-aroma type Baijiu from different regions using comprehensive two-dimensional gas chromatography–time-of-flight mass spectrometry and descriptive sensory analysis. Food Chem. 331, 127335. https://doi.org/10.1016/j.foodchem.2020.127335.

He, Z.L., Yang, K.Z., Liu, Z.P., An, M.Z., Qiao, Z.W., Zhao, D., Zheng, J., Qian, M.C., 2021. Tandem solid-phase extraction columns for simultaneous aroma extraction and fractionation of Wuliangye and other baijiu. Molecules 26 (19), 6030. https://doi.org/10.3390/molecules26196030.

Hong, J.X., Huang, H., Zhao, D.R., Sun, J.Y., Huang, M.Q., Sun, X.T., Sun, B.G., 2023. Investigation on the key factors associated with flavor quality in northern strong aroma type of Baijiu by flavor matrix. Food Chem. 426, 136576. https://doi.org/10.1016/j.foodchem.2023.136576.

Hong, J.X., Wang, J.S., Zhang, C.S., Zhao, Z.G., Tian, W.J., Wu, Y.S., Chen, H., Zhao, D. R., Sun, J.Y., 2021. Unraveling variation on the profile aroma compounds of strong aroma type of Baijiu in different regions by molecular matrix analysis and olfactory analysis. RSC Adv. 11 (53), 33511–33521. https://doi.org/10.1039/d1ra06073b.

Ickes, C.M., Cadwallader, K.R., 2017. Effects of ethanol on flavor perception in alcoholic beverages. Chemosens. Percept. 10, 119–134. https://doi.org/10.1007/s12078-017-9238-2.

Jin, G.Y., Zhu, Y., Xu, Y., 2017. Mystery behind Chinese liquor fermentation. Trends Food Sci. Technol. 63, 18–28. https://doi.org/10.1016/j.tifs.2017.02.016.

Lin, L.Y., Fan, W.L., Xu, Y., Zhu, D.C., Yang, T., Li, J., 2024. Characterization of key odorants in Chinese Texiang aroma and flavor type baijiu (Chinese liquor) by means of a molecular sensory science approach. J. Agric. Food Chem. 72 (2), 1256–1265. https://doi.org/10.1021/acs.jafc.3c07053.

Longo, R., Blackman, J.W., Torley, P.J., Rogiers, S.Y., Schmidtke, L.M., 2017. Changes in volatile composition and sensory attributes of wines during alcohol content reduction. J. Sci. Food Agric. 97 (1), 8–16. https://doi.org/10.1002/jsfa.7757.

Ma, Y., Beno, N., Tang, K., Li, Y., Simon, M., Xu, Y., Thomas-Danguin, T., 2022. Assessing the contribution of odor-active compounds in icewine considering odor mixture-induced interactions through gas chromatography-olfactometry and Olfactoscan. Food Chem. 388, 132991. https://doi.org/10.1016/j.foodchem.2022.132991.

Ma, Y., Guilbert, A., Beno, N., Tang, K., Xu, Y., Thomas-Danguin, T., 2023. Exploring the effects of mixture composition factors and perceptual interactions on the perception of icewine odor: An olfactometer-based study. Food Chem. 429, 136881. https://doi.org/10.1016/j.foodchem.2023.136881.

Ma, Y., Xu, Y., Tang, K., 2025. Olfactory perception complexity induced by key odorants perceptual interactions of alcoholic beverages: Wine as a focus case example. Food Chem. 463, 141433. https://doi.org/10.1016/j.foodchem.2024.141433.

Ou, X. (2025a). *Baijiu industry in China* - statistics & facts. Statista. Retrieved Feb 24, 2025 from ⟨https://www.statista.com/topics/7815/baijiu-industry-in-china/#topicOverview⟩.

Ou, X. (2025b). *Revenue breakdown of baijiu industry in China 2023, by aroma*. Statista. Retrieved Feb 24 from ⟨https://www.statista.com/statistics/1365334/china-breakdown-of-baijiu-revenue-by-flavor/#statisticContainer⟩.

Paluchová, J., Berčík, J., Horská, E., 2017. Sens. Aroma Mark. [Dr. Diss. Wagening. Univ.]. https://doi.org/10.3920/978-90-8686-841-4_2.

Peinado, R.A., Mauricio, J.C., Medina, M., Moreno, J.J., 2004. Effect of Schizosaccharomyces pombe on aromatic compounds in dry sherry wines containing high levels of gluconic acid. J. Agric. Food Chem. 52 (14), 4529–4534. https://doi.org/10.1021/jf049853r.

Pineau, B., Barbe, J.C., Van Leeuwen, C., Dubourdieu, D., 2007. Which impact for β-damascenone on red wines aroma? J. Agric. Food Chem. 55 (10), 4103–4108. https://doi.org/10.1021/jf070120r.

Qiao, L.N., Wang, J., Wang, R.F., Zhang, N., Zheng, F.P., 2023. A review on flavor of Baijiu and other world-renowned distilled liquors. Food Chem. X 20, 100870. https://doi.org/10.1016/j.fochx.2023.100870.

Song, X.B., Jing, S., Zhu, L., Ma, C.F., Song, T., Wu, J.H., Zhao, Q.Z., Zheng, F.P., Zhao, M.M., Chen, F., 2020. Untargeted and targeted metabolomics strategy for the classification of strong aroma-type baijiu (liquor) according to geographical origin using comprehensive two-dimensional gas chromatography-time-of-flight mass spectrometry. Food Chem. 314, 126098. https://doi.org/10.1016/j.foodchem.2019.126098.

Tan, Y.X., Siebert, K.J., 2004. Quantitative structure–activity relationship modeling of alcohol, ester, aldehyde, and ketone flavor thresholds in beer from molecular features. J. Agric. Food Chem. 52 (10), 3057–3064. https://doi.org/10.1021/jf035149j.

Taylor, A.J., 2023. Predicting sensory properties from chemical profiles, the ultimate flavour puzzle: a tale of interactions, receptors, mathematics and artificial intelligence. In: Lawlor, In.J.B., McEwan, J.A., Labbe, D. (Eds.), Digital Sensory Science. Elsevier, pp. 95–115.

Van Ruth, S.M., 2001. Methods for gas chromatography-olfactometry: a review. Biomol. Eng. 17 (4-5), 121–128. https://doi.org/10.1016/S1389-0344(01)00070-3.

Wang, J.S., Chen, H., Wu, Y.S., Zhao, D.R., 2022. Uncover the flavor code of strong-aroma baijiu: Research progress on the revelation of aroma compounds in strong-aroma baijiu by means of modern separation technology and molecular sensory evaluation. J. Food Compos. Anal. 109, 104499. https://doi.org/10.1016/j.jfca.2022.104499.

Wang, L., Wu, L., Xiang, D., Huang, H., Han, Y., Zhen, P., Shi, B., Chen, S., Xu, Y., 2023. Characterization of key aroma compounds in aged Qingxiangxing baijiu by comparative aroma extract dilution analysis, quantitative measurements, aroma recombination, and omission studies. Food Chem. 419, 136027. https://doi.org/10.1016/j.foodchem.2023.136027.

Xiangli, R., Ma, Y., Zeng, Y., Tang, K., Chen, S., Xu, Y., 2024. Differences and correlations between industrial experts and semi-trained assessors in the sensory evaluation of strong-aroma baijiu using rate-all-that-apply. J. Food Sci. 89 (9), 5841–5857. https://doi.org/10.1111/1750-3841.17280.

Zhao, D., Zheng, J., 2019. Research progress on aroma compounds in Wuliangye. In: Guthrie, B., Beauchamp, J.D., Buettner, A., Toth, S., Qian, M.C. (Eds.), Sex, smoke, and spirits: The role of chemistry. American Chemical Society, pp. 253–261.

Zhao, D.R., Shi, D.M., Sun, J.Y., Li, A.J., Sun, B.G., Zhao, M.M., Chen, F., Sun, X.T., Li, H. H., Huang, M.Q., 2018. Characterization of key aroma compounds in Gujinggong Chinese Baijiu by gas chromatography–olfactometry, quantitative measurements, and sensory evaluation. Food Res. Int. 105, 616–627. https://doi.org/10.1016/j.foodres.2017.11.074.

Zhao, T., Wu, J.Y., Meng, J.F., Shi, P.B., Fang, Y.L., Zhang, Z.W., Sun, X.Y., 2019. Harvesting at the right time: Maturity and its effects on the aromatic characteristics of Cabernet Sauvignon wine. Molecules 24 (15), 2777. https://doi.org/10.3390/molecules24152777.

Zhou, X., 2019. Analysis of volatile compounds of Strong-flavor base liquor. Study grade Recognit. [Master'S. Diss. Jiangsu Univ.